UFC

PRINCÍPIOS DE ELETRICIDADE E MAGNETISMO

CONSTANTES

EQUAÇÕES

TABELAS

TABELA DE INTEGRAIS

$$\int \frac{1}{\left(x^2+a^2\right)^{3/2}}dx \rightarrow \frac{1}{a^2}\frac{x}{\sqrt{x^2+a^2}}$$

$$\int \frac{x}{\left(x^2+a^2\right)^{3/2}}dx \rightarrow \frac{-1}{\sqrt{x^2+a^2}}$$

$$\int \frac{x^2}{\left(x^2+a^2\right)^{3/2}}dx \rightarrow \frac{-x}{\sqrt{x^2+a^2}} + \ln\left(x+\sqrt{x^2+a^2}\right)$$

$$\int \frac{1}{\left(x^2+a^2\right)^{1/2}}dx \rightarrow \ln\left(x+\sqrt{x^2+a^2}\right)$$

$$\int \frac{x}{\left(x^2+a^2\right)^{1/2}}dx \rightarrow \sqrt{x^2+a^2}$$

$$\int \frac{1}{\left(x^2+a^2\right)}dx \rightarrow \frac{1}{\sqrt{a^2}}\operatorname{atan}\left(\frac{x}{\sqrt{a^2}}\right)$$

$$\int x.dx \rightarrow \frac{x^2}{2}$$

$$\int \frac{1}{x}dx \rightarrow \ln(x)$$

$$\int \frac{1}{x^2}dx \rightarrow -\frac{1}{x}$$

$$\int \frac{1}{(a.x+b)}dx \rightarrow \frac{\ln(a.x+b)}{a}$$

$$\int \frac{1}{(a.x+b)^2}dx \rightarrow \frac{-1}{a.(a.x+b)}$$

CONSTANTES

$\pi = 3{,}141592654$

$c = 299.792.458$ [m/s] (velocidade da luz no vácuo)

$e = 1{,}6021766208 \ .10^{-19}$ [C] (carga elementar)

$\mu_0 = 4.\pi.10^{-7}$ [H/m] (permeabilidade magnética no vácuo)

$\varepsilon_0 = \dfrac{1}{\mu_0.c^2} = 8{,}85418782.10^{-12}$ [F/m] (permissividade elétrica no vácuo)

$\dfrac{1}{4.\pi.\varepsilon_0} = K = 8{,}98755179 \ .10^{9} \ \ [\dfrac{N.m^2}{C^2}]$

(Listagem com 4 algarismos significativos)

$\pi = 3{,}142$

$e = 2{,}718$ (neperiano)

$e = 1{,}602.10^{-19}$ [C]

$\varepsilon_0 = 8{,}854.10^{-12}$ [F/m]

$\mu_0 = 1{,}257.10^{-6}$ [H/m]

$K = 8{,}988.10^{9} \ \ [\dfrac{N.m^2}{C^2}]$

CONSTANTES / DADOS

$$mol = 6{,}022.10^{23}\ [\text{entidades}]$$

Cobre:

$$A_{cu} = 63{,}54\left[\frac{\text{g}}{\text{mol}}\right]$$

$$\mu_{cu} = 3{,}2.10^{-3}\left[\frac{\text{m}^2}{\text{V.s}}\right]$$

$$\rho_{cu} = 8{,}960.10^{3}\left[\frac{\text{kg}}{\text{m}^3}\right]$$

$$1\ polegada = 25{,}40\ mm$$

$$1\ \text{mil} = \frac{1\ \text{polegada}}{1000}$$

Alumínio:

$$\sigma_{al} = 35{,}38.10^{6}\left[\frac{\text{S}}{\text{m}}\right]\ @\ 20^0 C$$

$$\rho_{al} = 2{,}826.10^{-8}[\Omega.\text{m}]\ @\ 20^0 C$$

Cobre recozido:

$$\sigma_{cu} = 58{,}00.10^{6}\left[\frac{\text{S}}{\text{m}}\right]\ @\ 20^0 C$$

$$\rho_{cu} = 1{,}724.10^{-8}[\Omega.\text{m}]\ @\ 20^0 C$$

$$e^{-1} = 0{,}3679$$

$$1 - e^{-1} = 0{,}6321$$

$$1 - e^{-5} = 0{,}9933$$

$$\text{Melétron} = 9{,}109.10^{-31}\ [\text{kg}]$$

$$\text{Mpróton} = 1{,}673.10^{-27}\ [\text{kg}]$$

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